CHAPTER

23 Anaesthesia and pain relief

Learning objectives

To gain an understanding of:

- Techniques of anaesthesia and airway maintenance
- Methods of providing pain relief
- Local and regional anaesthesia techniques
- The management of chronic pain and pain from malignant disease

HISTORY

Anaesthesia, as we know it today, was first successfully demonstrated by William Morton, a local dentist, at the Massachusetts General Hospital, Boston, MA, USA on 16 October 1846 when he administered ether to Gilbert Abbot for an operation on a vascular tumour on his neck. Earlier Horace Wells had used nitrous oxide in 1844 for painless extraction of teeth successfully.

Simpson, at Edinburgh University, overcame some of the technical difficulties of ether administration by introducing chloroform. The benefits of anaesthesia were then universally recognised and antagonism by religious leaders was countered when Queen Victoria accepted chloroform from John Snow during the birth of Prince Leopold in 1853.

KEY PRINCIPLES OF ANAESTHESIA

Optimum patient care is dependent on a collaborative approach by the anaesthetic and surgical teams. The importance of multidisciplinary collaboration has been clearly demonstrated by national audits such as the National Confidential Enquiry into Patient Outcome and Death (NCEPOD) and the Confidential Enquiry into Maternal Deaths in the UK. These audits have led to changes in clinical and non-clinical practice to improve morbidity and mortality.

The use of a safety checklist in operating theatres in the form of the World Health Organization's (WHO) 'WHO Surgical Safety Checklist' has shown a reduction in the incidence of perioperative untoward events.

The role of the modern anaesthetist has evolved from just being responsible for the patient in the operating suite into a 'perioperative physician' who optimises the patient for surgery, assesses and minimises risk, cares for the patient during the operation and then manages both pain and homeostasis in the postoperative period.

Summary box 23.1

Ground rules for anaesthesia

- Safe surgery is achieved by close teamwork between the surgeon and the anaesthetist
- Safety checklists ensure that things are not forgotten
- Risk assessments allow the best strategy to be chosen
- Anaesthetists are extending their care into the pre- and postoperative phases

PREPARATION FOR ANAESTHESIA

A surgeon's role is to carry out, in cooperation with the anaesthetist, a thorough preoperative assessment that recognises medical and anaesthetic risk factors and facilitates the optimisation of the patient's condition (see ***Chapter 21***).

Anaesthesia; the name was suggested by Oliver Wendell-Homes, first appeared in Bailey's *An Universal Etymological English Dictionary* in 1751.
William Thomas Gren Morton, 1819–1868, dentist who practised in Boston, MA, USA.
Horace Wells, 1815–1848, Harvard, CT, USA, dentist who pioneered the use of nitrous oxide anaesthesia to prevent pain during dental procedures.
Sir James Young Simpson, 1811–1870, Professor of Midwifery, Edinburgh, UK.
John Snow, 1813–1858, general practitioner, London, UK, was one of the pioneers of anaesthesia.
Humphrey Davy, 1800, suggested that nitrous oxide inhalation might be used to relieve the pain of surgical operations and named it 'laughing gas'.
Henry Edmund Gaskin Boyle, in 1917, got his gas-oxygen machine, which became the first 'Boyle apparatus'.
The first examination for a **Diploma in Anaesthesia** was held in London in 1935.
The **First Chair in Anaesthesia**: Ralph Waters, Wisconsin, USA, in 1933 and RR Macintosh in Oxford, UK, in 1937.
During the First World War **Sir Ivan Magill** and **Stanley Rowbotham**, while working with Harold Gillies (pioneer of plastic surgery), developed tracheal intubation. Sir Magill is also remembered for his laryngoscope, Magill attachment and laryngeal forceps.

A careful preassessment, multidisciplinary approach and standardised care pathway with a carefully chosen anaesthetic and analgesic technique is the cornerstone of the 'enhanced recovery programmes' that have been introduced recently across the surgical specialties.

GENERAL ANAESTHESIA

General anaesthesia is commonly described as the triad of unconsciousness, analgesia and muscle relaxation.

Summary box 23.2

The general anaesthetic triad

- Amnesia: loss of awareness
- Analgesia: pain relief
- Muscle relaxation

Induction of general anaesthesia is most frequently done by intravenous agents. Propofol has replaced thiopentone as the most widely used induction agent and can be used for maintenance of anaesthesia. Other infrequently used intravenous agents include etomidate and ketamine. Newer agents based on a benzodiazepine receptor agonist, etomidate derivatives and fospropofol are still in the experimental stage.

Inhalational induction using agents such as non-pungent sevoflurane is useful in children, needle-phobic adults and those in whom a difficult airway is anticipated. These patients will have a higher risk of developing airway obstruction. *Figure 23.1* shows a commonly used anaesthetic machine.

Figure 23.1 Anaesthetic machine.

Summary box 23.3

Key features of commonly used intravenous anaesthetic agents

- **Propofol (di-isopropyl phenol)**: smooth induction, better haemodynamic stability, blunting of autonomic reflexes and ability to use as a continuous infusion
- **Thiopentone (barbiturate)**: rapid induction, myocardial depression. A reduced metabolic rate and lowering of intracranial pressure is useful in neurosurgical patients but the drop in blood pressure can have detrimental effects
- **Etomidate (steroid derivative)**: good haemodynamic stability, brief duration of action, but concern over adrenocortical depression
- **Ketamine (phencyclidine derivative)**: preservation of blood pressure and respiratory reflexes together with excellent analgesia makes it an ideal choice for field anaesthesia. Emergence delirium is associated with administration of ketamine

Rapid sequence induction (RSI) using a predetermined dose of intravenous anaesthetic agent together with a rapidly acting muscle relaxant is used in those with a high risk of regurgitation in order to secure the airway quickly. Commonly needed in emergency surgery, it is also a technique of choice in any non-emergency surgery in a patient with delayed emptying of the stomach.

Total intravenous anaesthesia (TIVA) is becoming popular following the introduction of propofol and the ultra-short-acting opioid remifentanil. The lack of cumulative effect, better haemodynamic stability, excellent recovery profile and concerns over environmental effects of inhalational agents have made TIVA an attractive choice. TIVA is routinely used in neurosurgery, airway laser surgery, during cardiopulmonary bypass and for day case anaesthesia.

Summary box 23.4

Special terms in anaesthesia

- RSI is a technique that allows the airway to be rapidly secured. It is used when there is a high risk of regurgitation that may lead to pulmonary aspiration
- TIVA is becoming increasingly popular

Maintenance of anaesthesia can be done using a continuous infusion of intravenous agent (propofol) or an inhaled vapour such as isoflurane, sevoflurane or desflurane.

The use of nitrous oxide is declining, despite its analgesic and weak anaesthetic properties, because of concerns over postoperative nausea and vomiting. It also increases the size

of air bubbles, causing adverse effects, for example, in eye, ear and abdominal surgery. Finally it is possibly mutagenic and is a powerful greenhouse gas.

Management of the airway during anaesthesia

Loss of muscle tone as a result of general anaesthesia means that the patient can no longer keep their airway open. Therefore, patients need their airway to be maintained for them. The use of muscle relaxants will mean that they will also be unable to breathe for themselves and so will require artificial ventilation. Head tilt, chin lift and jaw thrust manoeuvres along with adjuncts such as oropharyngeal airways (***Figure 23.2***) are used to facilitate bag–mask ventilation while induction agents exert their full effect. A laryngeal mask airway or endotracheal tube is then inserted, and the patient is allowed to breathe spontaneously or is ventilated during the procedure.

The addition of a cuff to the endo-tracheal tube facilitates positive pressure ventilation and protects the lungs from aspiration of regurgitated gastric contents.

Supraglottic airways

- **Laryngeal mask airway (LMA)**. Developed by Dr Archie Brain in the UK, the original LMA® is a first-generation supraglottic airway. The mask with an inflatable cuff is inserted via the mouth and produces a seal around the glottic opening, providing a very reliable means of maintaining the airway. Its placement is less irritating and less traumatic to a patient's airway than endotracheal intubation. The technique can be easily taught to non-anaesthetists and paramedics and can be used as an emergency airway management tool. Several varieties of first-generation LMAs are available, including the classic LMA and the flexible LMA. Further advancements have led to the development of second-generation supraglottic devices, such as the ProSeal® LMA and the i-gel® (***Figure 23.3***). These devices usually have an in-built 'bite block' and oesophageal drain tube. They can be used for ventilation of the lungs at higher inflation pressures and are more suitable for patients with a higher body mass index. There are also modified versions of the LMA, including the ILMA (intubating LMA), that allow a blind technique, aiding insertion of a tracheal tube in difficult conditions. There is increasing evidence that second-generation devices have a good safety and efficacy profile and should be replacing all first-generation devices.
- **Difficult intubation**. Endotracheal intubation is feasible in most patients, but in a certain proportion of patients it may be difficult or impossible. However, if it is compounded by an inability to ventilate and therefore maintain oxygenation of the patient by bag–mask, the consequence can be catastrophic hypoxia. Many devices have been developed to aid intubation if difficulty is anticipated (e.g. McGrath® blade, Airtraq®, C-MAC® video laryngoscope) (***Figures 23.4 and 23.5***); similarly, protocols have been created by specialised societies to deal

Figure 23.2 Oropharyngeal airways.

Figure 23.3 i-gel® supraglottic airway.

Figure 23.4 Single-use laryngoscope.

Archie Brain, b. 1942, formally an anaesthetist, whose patent application for the laryngeal mask airway was granted in 1982.

Figure 23.5 Airtraq® intubating device.

Figure 23.6 Fibreoptic intubating bronchoscope.

with such situations. One specialised method for intubation in difficult situations is the use of the fibreoptic intubating bronchoscope (***Figure 23.6***), facilitated by topical local anaesthetic in awake patients or using general anaesthesia. The anaesthetist places the endotracheal tube in the trachea by threading the tube over the bronchoscope, and so places the tube in the trachea under direct bronchoscopic vision. An awake intubation requires careful patient selection, as it may not be a suitable technique for all patient groups.

Double-lumen tubes and endobronchial tubes are used in procedures such as thoracoscopic, pulmonary and oesophageal surgery to allow collapse of one lung (while ventilating the other) for ease of surgery. Their use is also essential to isolate the healthy lung in empyema and in the case of a bronchopleural fistula.

Ventilating bronchoscopes and endobronchial catheters can be used to maintain oxygenation during laryngotracheal surgery or bronchoscopy by using intermittent jets of oxygen.

Summary box 23.5

Techniques for maintaining an airway

- Chin lift and jaw thrust: suitable for the short term when no aid is available
- Guedel airway: holds the tongue forward but does not prevent aspiration
- Supraglottic device: easy insertion, reliable airway, allows ventilation
- Endotracheal intubation: secure and protected airway

Summary box 23.6

Complications of intubation

- Failed intubation
- Accidental bronchial intubation
- Trauma to teeth, pharynx or larynx
- Aspiration of gastric contents during intubation
- Disconnection, blockage or kinking of tube
- Delayed tracheal stenosis

Muscle relaxation and artificial ventilation

Pharmacological blockade of neuromuscular transmission by neuromuscular blocking agents provides relaxation of muscles, allowing easy surgical access. However, the patient will need artificial ventilation. Neuromuscular blocking agents are broadly classified into depolarising and non-depolarising groups according to their mode of action (***Table 23.1***).

Suxamethonium is the most commonly used depolarising agent. It binds to the nicotinic acetylcholine receptors, resulting in opening of the cation channel, leading to depolarisation and rapid relaxation of muscles. Despite its adverse effects such as hyperkalaemia, muscle pain, anaphylaxis and potentially life-threatening malignant hyperthermia, suxamethonium is still widely used because of its quick onset and short duration of action. These properties are useful when rapid endotracheal intubation is necessary to protect the patient's airway or when short duration surgery is performed.

Non-depolarising muscle relaxants act by competitive blockade of postsynaptic receptors at the neuromuscular junction. They provide longer, predictable activity but require careful monitoring, appropriate timing and reversal of their action by agents such as neostigmine and sugammadex at the end of the procedure. A peripheral nerve stimulator is routinely used to monitor the depth of neuromuscular block and also to confirm satisfactory recovery of muscle power prior to extubation. With the increasing availability and evidence of the use of sugammadex, the non-depolarising muscle relaxant rocuronium is an alternative to suxamethonium in the 'rapid-sequence' induction as it allows reversal of its actions with sugammadex in a rapid manner.

Arthur Ernest Guedel, 1883–1956, Clinical Professor of Anesthesiology, University of Southern California, Los Angeles, CA, USA.

TABLE 23.1 Properties of commonly used muscle relaxants

	Advantages	Disadvantages
Suxamethonium	• Quickest onset, very short duration, spontaneous recovery • Ideal for rapid intubation and for short procedures	• Muscle pain, hyperkalaemia, prolonged apnoea and life-threatening malignant hyperthermia
Vecuronium	• Long acting • Minimal cardiovascular effect and less allergic reaction	• Dependent on hepatic metabolism and renal clearance; hence, caution if hepatic and renal impairment
Atracurium	• Intermediate acting • Non-enzymatic Hofmann degradation • Suitable in renal and hepatic failure	• Histamine release and allergic reactions
Rocuronium	• Rapid onset, intermediate action • Suitable for rapid intubation • Rapid reversal possible using sugammadex	• Allergic reactions • Excreted unchanged via bile and urine

Ventilation during anaesthesia

Mechanical ventilation is required when the patient's spontaneous ventilation is inadequate or when the patient is not breathing because of the effects of the anaesthetic, analgesic agents or muscle relaxants.

In volume control ventilation, a preset volume is delivered by the machine irrespective of the airway pressure. The pressure generated will be, in part, dependent on the resistance and compliance of the airway. In laparoscopic surgery requiring the Trendelenburg position (the patient is positioned head down), morbidly obese patients and those with lung disease, this may result in excessive pressures being developed, which may lead to barotrauma (pneumothorax).

In pressure control mode the ventilator generates flow until a preset pressure is reached. The actual tidal volume delivered is variable and depends on airway resistance, intra-abdominal pressure and the degree of relaxation.

Positive end-expiratory pressure (PEEP) is often applied to help maintain functional residual capacity. This avoids lung collapse by opening collapsed alveoli and maintains a greater area of gas exchange, so reducing vascular shunting.

Monitoring and care during anaesthesia

A minimum basic monitoring of cardiovascular parameters is required during surgery. This includes:

- vascular
  - electrocardiogram (ECG);
  - blood pressure;
- adequacy of ventilation:
  - inspired oxygen concentration;
  - oxygen saturation by pulse oximetry;
  - end-tidal carbon dioxide concentration.

Monitors of temperature, ventilation parameters and delivery of anaesthetic agents are also routinely used, while measurement of urine output and central venous pressure are recommended for major surgery.

Summary box 23.7

Intermittent positive-pressure ventilation

- Volume controlled, which ensures adequate gas entry but risks high-pressure damage
- Pressure controlled, which avoids high-pressure damage but risks inadequate ventilation
- PEEP reduces alveolar collapse and reduces vascular shunting so improving perfusion

LOCAL ANAESTHESIA

Local anaesthetic drugs (*Table 23.2*) may be used to provide anaesthesia and analgesia, as a sole agent or as adjuncts to general anaesthesia. Available techniques include topical anaesthesia, local infiltration, regional nerve blocks and central neuraxial blocks (spinal and epidural anaesthesia).

TABLE 23.2 The common local anaesthetic drugs.

Name	Maximum dose	Comments
Lidocaine	3 mg/kg (7 mg/kg with adrenaline [epinephrine])	Early onset, short acting, good sensory block
Bupivacaine	2 mg/kg	Long lasting, more cardiotoxic, must never be used intravenously
Prilocaine	6 mg/kg (9 mg/kg with adrenaline)	Least systemic toxicity, causes methaemoglobinaemia
Ropivacaine	3–4 mg/kg	Less cardiotoxic, greater sensory–motor separation
Levobupivacaine	2 mg/kg	Isomer of bupivacaine with fewer cardiotoxic properties

Local anaesthesia techniques can lead to complications that may be local, such as infection or haematoma, or systemic, as a result of overdose or accidental intravascular injection.

Friedrich Trendelenburg, 1844–1924, successively Professor of Surgery at Rostock (1875–1882), Bonn (1822–1895) and Leipzig (1895–1911), Germany. The Trendelenburg position was first described in 1885.

The systemic effects of local anaesthetic agents are dose dependent and manifest as cardiovascular (cardiac arrhythmia, cardiac arrest) or neurological (depressed consciousness, convulsions). Prilocaine overdose causes methaemoglobinaemia, whereas bupivacaine overdose causes treatment-resistant ventricular arrhythmia and cardiac arrest.

The addition of adrenaline (epinephrine) to local anaesthetic solutions hastens onset, prolongs the duration of action and permits a higher upper dose limit. The use of adrenaline is contraindicated in patients with cardiovascular disease, those taking tricyclic and monoamine oxidase inhibitors and in end-arterial locations.

Appropriately skilled personnel, resuscitation equipment and oxygen should always be available with local anaesthetic use because of the potential risks of life-threatening complications.

Regional anaesthesia

Regional anaesthesia involves central neuraxial or peripheral nerve or plexus blocks using local anaesthetic drugs. It has a clear advantage when general anaesthesia carries a higher risk of morbidity and mortality, such as in patients with debilitating respiratory and cardiovascular disease and obstetric cases. It also provides excellent pain relief in the postoperative period, reducing the need for analgesics such as opioids.

As with general anaesthesia, venous access should be established and vital parameters should be monitored during regional anaesthesia.

Localising nerves using anatomical landmarks and eliciting paraesthesia alone carries a high risk of nerve damage and intravascular injection and has a lower success rate. The use of nerve stimulators to localise nerves improves the success rate and reduces risks. Ultrasound-guided regional anaesthesia allows the visualisation of nerves and the spread of local anaesthetics, enabling the use of a smaller dose of local anaesthetic agents with improved success rates and safety.

Summary box 23.8

Types of anaesthesia

- General anaesthesia may be more acceptable to patients
- Regional anaesthesia has major advantages in obstetrics and patients with respiratory compromise
- Local blocks have been transformed by nerve stimulators and ultrasound guidance
- All require full resuscitation and monitoring equipment to be available

Common local anaesthesia techniques

Topical anaesthesia

- **EMLA (eutectic mixture of local anaesthetics)**. This is a mixture of lidocaine and prilocaine for application to the skin for venepuncture in children.
- **Cocaine**. It may be called Moffett's solution (with an added mixture of adrenaline and sodium bicarbonate) and used in nasal surgery for anaesthesia and vasoconstriction.
- **Lidocaine 2/4/10%**. Spray to anaesthetise the airway during awake fibreoptic intubation.

Nerve blocks

An interscalene approach to brachial plexus for shoulder surgery produces excellent postoperative analgesia. Complications include phrenic nerve block, Horner's syndrome and accidental intravascular and spinal injection. Supraclavicular, infraclavicular or axillary approaches to the brachial plexus can be used as the sole anaesthetic technique for upper limb surgery (*Figure 23.7*). Femoral and sciatic nerve blocks are often used for anaesthesia and analgesia for lower limb surgery.

Local anaesthetic blocks have been described that block the nerves closer to the joint so that motor blockade of the whole lower limb can be minimised. An example of this type of block is the iPACK (infiltration between the popliteal artery and capsule of the knee) block, which targets the genicular nerves supplying the knee joint by using ultrasound-guided injection posterior to the knee joint.

Transversus abdominis plane/quadratus lumborum and erector spinae plane block are examples of fascial plane blocks, which have grown rapidly in popularity. Local anaesthetic is injected in the fascial planes containing the nerves from abdominal or chest wall structures. Transversus abdominis plane (TAP) block has been claimed to provide effective

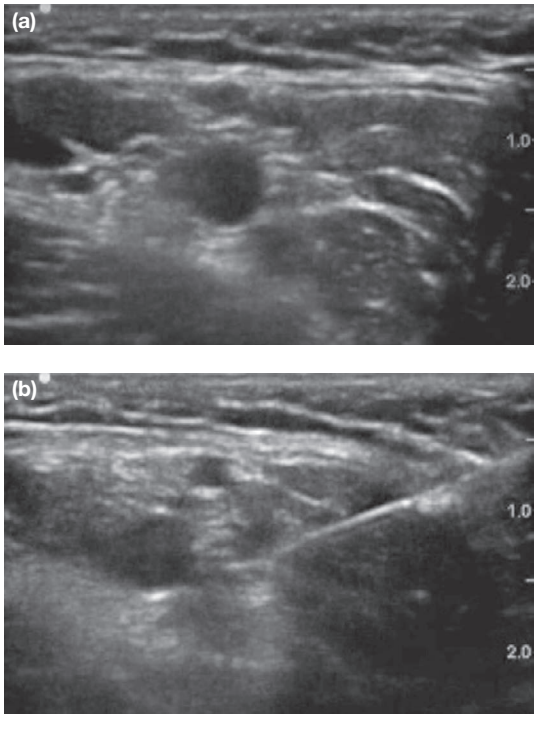

Figure 23.7 (a, b) Ultrasonic picture of brachial plexus block.

Johann Friedrich Horner, 1831–1886, Professor of Ophthalmology, Zurich, Switzerland, described this syndrome in 1869.
Major A. J. Moffett of the Royal Army Medical Corps first described its use in 1941. He modified it in 1946 to include 2 mL of 8% cocaine, 2 mL of 1% sodium bicarbonate, and 1 mL of 1:1000 adrenaline.

analgesia after a wide range of abdominal surgery. The T6–L1 segmental nerves enter the plane between the internal oblique muscle and the transverse abdominis muscle just medial to the anterior axillary line. Injection of local anaesthetic into the fascial plane between the internal oblique and transversus abdominis muscles allows a block of all these nerves, and excellent anaesthesia of the anterior abdominal wall. Quadratus lumborum block also aims to target abdominal nerves as they pass in front of the quadratus lumborum muscle. Erector spinae block aims to block the spinal nerves at various sites, depending on the site of injection. Local anaesthetic is deposited underneath the erector spinae group of muscles as they lie over the transverse process of the vertebra.

Intravenous regional anaesthesia (Bier's block)

Bier's block produces excellent anaesthesia for short surgery, particularly for the upper limb (e.g. carpal tunnel release). In this technique local anaesthetic is injected intravenously to produce anaesthesia of the upper limb. A double tourniquet is used on the side of surgery. An intravenous cannula is sited into a vein on the back of the hand on the side that is being operated on. The upper limb is then exsanguinated using an Esmarch bandage. The proximal cuff of the double tourniquet is inflated, followed by intravenous injection of prilocaine into the cannula. After 20 minutes the distal cuff of the tourniquet is inflated and then the proximal cuff is deflated. Even if surgery is finished, the tourniquet should be left inflated until the local anaesthetic has bound to tissues (20 minutes) so that release of local anaesthetic into the systemic circulation does not occur. Lidocaine can be used with caution (consider the safe dose of lidocaine and the time of tourniquet inflation) but bupivacaine should never be used for Bier's block.

Spinal anaesthesia

Spinal anaesthesia alone or in combination with general anaesthesia or sedation is used extensively for lower limb, obstetric and pelvic surgery. Injection of a 'single-shot' local anaesthetic agent intrathecally produces intense and rapid block for surgery. The addition of opioids provides prolonged postoperative analgesia but carries the risk of late respiratory depression.

Autonomic sympathetic blockade produces hypotension, particularly if the level of block is above the T10 spinal dermatome. Caution is needed in patients with hypovolaemia and cardiovascular disease.

The incidence of dural puncture headache can be minimised by limiting the number of punctures and using fine-bore pencil-tip needles that are designed to split rather than cut the dura.

Epidural anaesthesia

Epidural anaesthesia is slower in onset than spinal but has the advantage of prolonged analgesia by multiple dosing or continuous infusion through a catheter placed in the epidural space. Being slower in onset, the resulting hypotension from sympathetic blockade can be better controlled and can reduce blood loss.

Continuous infusion (with a patient-controlled bolus) of weak local anaesthetic combined with opioids (such as fentanyl) is routinely used for postoperative analgesia. Placement of an epidural catheter in the high thoracic region provides excellent analgesia for a wide variety of upper abdominal and thoracic surgical operations, enabling early mobilisation and reducing respiratory complications.

Epidural anaesthesia is technically more difficult than spinal anaesthesia; it has a higher failure rate and carries the risk of nerve damage, spinal injuries, accidental spinal injection of a large volume of local anaesthetic, infection and epidural haematoma.

Summary box 23.9

Local anaesthetics

- EMLA cream for children needing injections
- Regional and nerve blocks for limb surgery
- Spinal anaesthesia offers a quick onset and a short duration of anaesthesia
- Epidurals are more difficult but can then be topped up postoperatively and used as a continuous infusion

PAIN

Pain is defined as 'An unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage.' Most patients will experience pain after surgery. This is usually managed by a combination of painkillers and local anaesthetic techniques (multimodal analgesia). The common painkillers that are used are regular paracetamol, non-steroidal anti-inflammatory drugs (NSAIDs) and opioids. Opioids can be weak, such as codeine, or stronger opioids, such as morphine and oxycodone. Opioids are associated with a high incidence of side effects (nausea and vomiting, respiratory depression, itching). By combining opioids with other drugs, the dose of opioid can be minimised, thus side effects can be reduced.

Paracetamol and non-steroidal anti-inflammatory drugs

Paracetamol was first synthesised in 1878 by Morse and was introduced for medical usage in 1883. However, because of misinterpretation of its safety profile, its use was limited until the 1950s, when the chemically similar, and until then preferred analgesic, phenacetin was withdrawn owing to renal toxicity. Paracetamol is probably now the most commonly used drug worldwide; it is available over the counter, used in almost all ages and forms step 1 of the WHO analgesic ladder. It is first-line treatment for pyrexia and pain, plays an important role in multimodal analgesia and is considered to possess a generally excellent safety profile, except in significant overdose, with only a few drug interactions. Oral and rectal administration can produce analgesia within 40 minutes, with maximal effect at

August Karl Gustav Bier, 1861–1949, Professor of Surgery, Bonn (1903–1907) and Berlin (1907–1932), Germany.

1 hour, but because of variability in bioavailabilities (in the range 63–89% for oral and 24–98% for rectally administered preparations) the timing of onset can be unpredictable. The introduction of its intravenously administered preparation within the last decade overcomes this issue.

NSAIDs are used in the treatment of acute pain for their opioid-sparing effects, as part of a multimodal analgesic regimen. However, it is important to recognise that long-term usage and an increase in prescription may be associated with significant morbidities. Increased risk of perioperative bleeding, gastrointestinal bleeding and ulceration, thrombotic events such as myocardial infarction and stroke, renal impairment, fluid retention and exacerbation of asthma are some of the side effects of NSAIDs, suggesting cautious usage.

Intravenous opioids administered as patient-controlled analgesia (PCA) for pain relief is a useful technique. The patient is trained to give a bolus dose of drug by pressing a control button on a machine, the functions of which have been regulated by medical staff. The strength, frequency and total dose of drug in a given time are all limited by computer. This method is popular with patients as they have control and prevents delays in administration of doses.

Chronic pain

Chronic pain is defined as pain that persists or recurs for more than 3 months. In chronic pain syndromes, pain can be the sole or a leading complaint and requires special treatment and care. This could be 'chronic primary pain', which may be conceived of as a disease in its own right, for example in conditions such as fibromyalgia or non-specific low-back pain. In six other subgroups, pain is secondary to an underlying disease: chronic cancer-related pain, chronic neuropathic pain, chronic secondary visceral pain, chronic post-traumatic and postsurgical pain, chronic secondary headache and orofacial pain and chronic secondary musculoskeletal pain. These conditions are summarised as 'chronic secondary pain', in which pain may at least initially be conceived as a symptom.

In surgical practice, patients with chronic pain may present for treatment of the cause (e.g. pancreatitis, malignancy) or concomitant benign pathology. Almost 50 years ago, using gate theory, Melzeck and Wall proposed pain to be not only subjective but also a multidimensional experience that incorporates sensory/discriminative, motivational/affect and cognitive aspects. Recent guidance from the International Association for the Study of Pain highlights that the multidimensional aspect is one of the key components, hence the treatment of chronic pain involves a multidisciplinary multimodal approach.

Mechanistically, chronic pain may be classified into:

- **Nociceptive pain**, which may result from musculoskeletal disorders or cancer activating cutaneous nociceptors (pain receptors). Prolonged ischaemic or inflammatory processes result in sensitisation of peripheral nociceptors and altered activity in the central nervous system, leading to exaggerated responses in the dorsal horn of the spinal cord. The widened area of hyperalgesia and increased sensitivity (allodynia) have been attributed to increased transmission in the central nervous system.
- **Neuropathic (or neurogenic) pain** is dysfunction in the peripheral or central nerves (excluding the 'physiological' pain due to noxious stimulation of the nerve terminals). It is classically of a 'burning', 'shooting' or 'stabbing' type and may be associated with allodynia, numbness and diminished thermal sensation. It is poorly responsive to opioids. Examples include trigeminal neuralgia, postherpetic neuropathy and diabetic neuropathy.

Principles of chronic pain management

Non-pharmacological treatment

This involves a multidisciplinary approach targeting the biopsychosocial model of health (***Figure 23.8***). Early assessment and engagement with physiotherapy and exercise are key contributors in successful management. Psychological interventions such as counselling, cognitive behaviour therapy and mindfulness are recognised strategies for pain control. 'Pain management programmes' lay out a logical structure for this.

Pharmacological treatment

Drugs in chronic non-malignant pain

Paracetamol and NSAIDs are the mainstays of musculoskeletal pain treatment. The tricyclic antidepressant drugs and anticonvulsant agents are often useful for the pain of nerve injury, although side effects can prove troublesome and reduce compliance. Both pregabalin and gabapentin reduce spontaneous neuronal activity by their action on the $\alpha_2\delta$ subunit of the calcium channel. They are now routinely used for managing neuropathic chronic pain. In more severe and debilitating non-malignant chronic pain, opioid analgesic drugs are used in slow release oral preparations (morphine and oxycodone) and transcutaneous patches (fentanyl and buprenorphine).

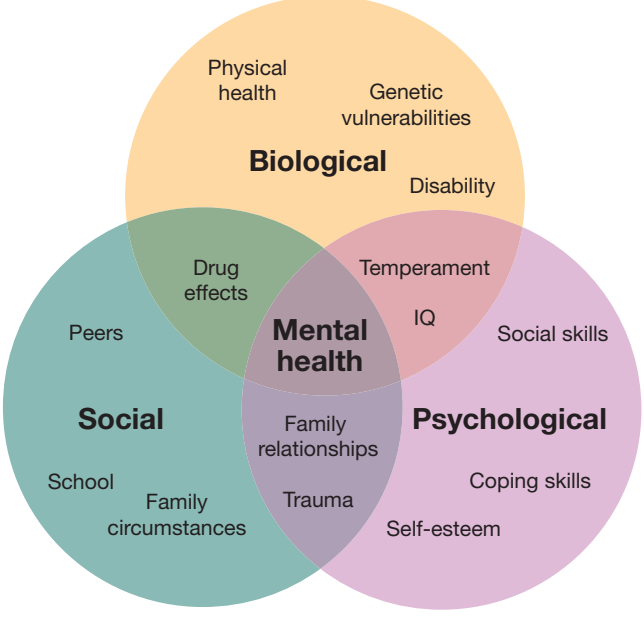

Figure 23.8 Biopsychosocial model of health.

Summary box 23.10

Commonly used terms in chronic pain

Term	Definition
Allodynia	Pain due to a stimulus that does not normally provoke pain. Allodynia involves a change in the quality of a sensation, whether tactile, thermal or of any other sort
Analgesia	Absence of pain in response to stimulation that would normally be painful
Central sensitisation	Increased responsiveness of nociceptive neurones in the central nervous system to their normal or subthreshold afferent input. The net effect is that innocuous stimuli will be interpreted as painful
Epidural space	The space (or potential space) between the ligamentum flavum (or vertebral wall) and the dura, just outside the spinal canal, extending from the foramen magnum to the sacrum. Leads and catheters may be placed in this space via a needle
Hyperalgesia	Abnormally heightened sensitivity to pain. For pain evoked by stimuli that usually are not painful, the term allodynia is preferred, while hyperalgesia is more appropriately used for cases with an increased response at a normal threshold or at an increased threshold
Hypoalgesia	Decreased sensitivity to painful stimuli. Hypoalgesia can be caused by exogenous chemicals such as opioids, as well as by chemicals produced by the body in phenomena such as fear- and exercise-induced hypoalgesia
Intrathecal space	The cerebrospinal fluid-filled space around the spinal cord, protected by the dura mater, into which certain medicines may be delivered to achieve their most potent effect
Nociception	Sensory response to certain harmful or potentially harmful stimuli. Nociception triggers a variety of biological and behavioural responses and may also result in a subjective experience of pain
Neuropathic pain	Neuropathic pain is defined as pain caused by a lesion or disease of the somatosensory nervous system
Nociceptive pain	Pain that arises from actual or threatened damage to non-neural tissue and that is due to the activation of nociceptors
Noxious stimulus	An actual or potential tissue-damaging event
Off-label	When a prescription drug is prescribed for uses other than what Australia's Therapeutic Goods Administration, the USA's Food and Drug Administration or the UK's Medicines and Healthcare products Regulatory Agency has approved and published in the drug's package insert
Pain	An unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage
Pain threshold	The minimum amount of a stimulus with which pain begins to be felt. It is an entirely subjective phenomenon
Paraesthesia	Abnormal cutaneous sensations such as tingling, tickling, pricking or burning with no apparent physical cause. Often described as 'pins and needles' or of a limb 'falling asleep'
Peripheral neuropathic pain	Pain caused by a lesion or disease of the peripheral somatosensory nervous system

Tapentadol, with its dual action on the opioid and noradrenaline (norepinephrine) selective reuptake inhibition pathway, may provide relief in patients with both a neuropathic and nociceptive element to their pain. Combinations of drugs often prove useful to achieve the optimum of efficacy with minimal side effects.

Treatment of pain dependent on sympathetic nervous system activity

Even minor trauma and surgery (often of a limb) can provoke excessive sympathetic adrenergic activity, inducing vasoconstriction and abnormal nociceptive transmission. This can lead to chronic burning pain, allodynia, trophic changes and resultant disuse.

Management includes antineuropathic pain medications (pregabalin, gabapentin, amitriptyline) as part of multimodal analgesia with a multidisciplinary pain management approach. This includes considerable input from psychological services and targeted physiotherapy. Interventional treatment may include local anaesthetic injection of the stellate ganglion for upper limb symptoms. Percutaneous chemical lumbar sympathectomy with local anaesthetic is used for relief of rest pain in advanced ischaemic disease of the legs.

Interventional pain management for chronic pain

Local anaesthetic and steroid injections can be effective around an inflamed nerve and they reduce the cycle of constant pain transmission with consequent muscle spasm. Transforaminal

selective root blocks in the epidural space are used for the pain of nerve root irritation associated with or without minor disc prolapse, followed by active physiotherapy and rehabilitation to promote mobility.

Nerve stimulation procedures such as acupuncture and transcutaneous nerve stimulation increase endorphin production in the central nervous system. Nerve decompression craniotomy rather than percutaneous coagulation of the ganglion is now performed for trigeminal neuralgia.

Spinal cord stimulation (SCS) by dorsal column stimulation is now a recognised and effective management of intractable neuropathic pain (***Figure 23.9***). This involves placement of electrodes in the posterior epidural space to allow dorsal column stimulation through an implantable pulse generator inserted in the body. In the UK this has been recommended by National Institute for Health and Care Excellence (NICE) guidance for intractable neuropathic pain management. Robust evidence exists for its clinical and health economic benefit in failed back surgery syndrome, in which patients have undergone previous spinal surgery. Early use of SCS in surgery-naive patients might lead to better outcomes. National guidelines in the UK do not recommend disc replacements or spinal fusions in patients with lower back pain unless the latter are part of a trial, rendering a large part of the patient population ineligible for surgery. In these instances, SCS could prove to be an effective solution.

Pain control in malignant disease

Pain is a common symptom associated with cancer, even more so during the advanced stages. In intractable pain, the underlying principle of treatment is to encourage independence of the patient and an active life in spite of the symptom. WHO advises use of the 'WHO analgesic ladder':

- first step: simple analgesics – paracetamol, NSAIDs, tricyclic drugs or anticonvulsant drugs;
- second step: intermediate-strength opioids – codeine, tramadol;
- third step: strong opioids – morphine (pethidine has now been withdrawn).

Oral opiate analgesia is necessary when the less powerful analgesic agents no longer control pain on movement or enable the patient to sleep. Opioids may exhibit both dependence and addiction with long-term use. It is important to distinguish between addiction and dependence; the former is a psychosocial phenomenon whereas the latter is a purely physiological response to a given drug. Some patients experience 'breakthrough pain' (acute, excruciating and incapacitating), which occurs either spontaneously or in relation to a specific predictable or unpredictable trigger experienced by patients who have relatively stable and adequately controlled background pain. Opioid rotation or switching may be considered if a patient obtains pain relief with one opioid and has severe adverse effects.

Oral morphine, which is often used for chronic pain, can be prescribed in short-acting liquid or tablet form and should be administered regularly every 4 hours until an adequate dose of drug has been titrated to control the pain over 24 hours. Once this is established, the daily dose can be divided into two separate administrations of enteric-coated, slow-release morphine tablets (MST morphine) every 12 hours. Additional short-acting opioids (morphine/fentanyl) can then be used to cover episodes of 'breakthrough pain'. Nausea treated using antiemetic agents does not usually persist, but constipation is a frequent and persistent complication requiring regular prevention by laxatives.

Figure 23.9 Dual-lead spinal cord stimulator in the epidural space.

Infusion of subcutaneous, intravenous, intrathecal or epidural opiate drugs

The infusion of an opiate is necessary if a patient is unable to take oral drugs. Subcutaneous infusion of diamorphine is effective and simple to administer. Epidural infusions of diamorphine with an external pump can be used in mobile patients. Intrathecal infusions with pumps programmed by external computers are used; however, there is a possibility of the patient developing an infection with catastrophic effects. Intravenous narcotic agents may be reserved for acute crises, such as pathological fractures.

Neurolytic techniques in cancer pain

These should only be used if life expectancy is limited and the diagnosis is certain. The useful procedures are:

- subcostal phenol injection for a rib metastasis;
- coeliac plexus neurolytic block with alcohol for the pain caused by pancreatic, gastric or hepatic cancer;
- intrathecal neurolytic injection of hyperbaric phenol;

- percutaneous anterolateral cordotomy to divide the spinothalamic ascending pain pathway; this is a highly effective technique in experienced hands, selectively eliminating pain and temperature sensation in a specific limited area.

Alternative strategies include:

- the development of anti-pituitary hormone drugs, such as tamoxifen and cyproterone, has enabled effective pharmacological therapy for the pain of widespread metastases instead of pituitary ablation surgery;
- palliative radiotherapy can be most beneficial for the relief of pain in metastatic disease; adjuvant drugs such as corticosteroids to reduce cerebral oedema or inflammation around a tumour, which may be useful in symptom control; tricyclic antidepressants, anticonvulsants and flecainide are also used to reduce the pain of nerve injury.

Summary box 23.11

Approximate equianalgesic potencies of opioids for oral administration

	Potency	Equivalent dose to 10 mg oral morphine
Codeine phosphate	0.1	100 mg
Dihydrocodeine	0.1	100 mg
Hydromorphone	5	2 mg
Morphine	1	10 mg
Oxycodone	1.5	6.6 mg
Tapentadol	0.4	100 mg

Adapted from the British National Formulary.

FURTHER READING

Chou R, Gordon DB, de Leon-Casasola OA *et al*. Management of postoperative pain: a clinical practice guideline from the American Pain Society, the American Society of Regional Anesthesia and Pain Medicine, and the American Society of Anesthesiologists' Committee on Regional Anesthesia, Executive Committee, and Administrative Council. *J Pain* 2016; **17**(2): 131–57.

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